Lab Assignment 2: Regression: Housing Price Prediction

CSAI2017P: Applied Machine Learning (Lab)

Experiment 3 (Unit V) • CO2, CO4

Instructor: Dr. Tofik Ali **Due:** announced in the lab

Student Name: ____________________ **SAP ID:** ______________

Batch: __________ **Lab quiz:** LQ2, after submission

Objective

- Fit simple and multiple linear regression and read the coefficients.
- Evaluate a regressor with metrics that carry the units of the target.
- Recognise over-fitting and control it with regularization.
- **Before the lab:** run `fetch_california_housing()` once at home. It downloads ~1.4 MB and the campus network may block it.

Datasets used in this assignment

A1 (California Housing).
Full descriptions, sources and loading snippets for all anchor datasets are in `Anchor-Datasets.pdf`, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA02_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			**10**

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 $(2 \times 2 = 4)$

A1. Simple linear regression **[2 marks]**
Dataset: A1. Target: `MedHouseVal`.

(a) Fit a regression of `MedHouseVal` on `MedInc` alone.

(b) Report the slope, the intercept and test R^2.

(c) State in one sentence what the slope means in real units.

A2. Multiple linear regression **[2 marks]**
Dataset: A1, all features.

(a) Fit a multiple linear regression on all eight features inside a scaling pipeline.

(b) Report test MAE, RMSE and R^2.

(c) Plot predicted against actual and describe the pattern in two lines.

A3. Metric choice **[2 marks]**
Dataset: A1.

(a) Compute MAE, MSE and RMSE for your model from A2.

(b) State which of the three is hardest to explain to a non-technical client, and why.

(c) State which you would report if a few very expensive districts matter most, and why.

Section B (Medium) — attempt any 1 of 2 $(1 \times 4 = 4)$

B1. Polynomial features and over-fitting **[4 marks]**
Dataset: A1.

(a) Build pipelines with `PolynomialFeatures` of degree 1, 2 and 3 followed by `LinearRegression`.

(b) Report train and test RMSE for each degree in one table.

(c) Identify the degree at which over-fitting begins and explain in 4–6 lines how the table shows it.

B2. Ridge and Lasso **[4 marks]**
Dataset: A1 with degree-2 polynomial features.

(a) Fit Ridge and Lasso for $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ and plot test RMSE against α.

(b) Report the best α for each and the count of exactly-zero coefficients in the best Lasso.

(c) Explain in 4–6 lines why Lasso zeroes coefficients and Ridge does not.

Section C (Hard) — compulsory (2)

C1. Where the model fails **[2 marks]**
Dataset: A1.

(a) Take the 20 test districts with the largest absolute error from your best model.

(b) Describe in 4–6 lines what they have in common and what you would add to the data to fix it.

End of Lab Assignment 2

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.